

RapidBoxForest: Cross-Revision Endpoint-Envelope Reuse for Region-Graph Motion Planning

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Abstract—Repeated-query motion planning in changing workcells requires geometric reuse without carrying obsolete collision claims across revisions. RapidBoxForest (RBF) stores obstacle-independent endpoint envelopes in a Lifelong Envelope Cache Tree (LECT), keyed by canonical dyadic addresses shared with region-graph primitives. Across compatible revisions, exact-key replay avoids rematching, while link envelopes, obstacle labels, and adjacency are rebuilt for the current scene. A budgeted builder refines cells, coalesces exact dyadic unions, and constructs corridors from current-scene witnesses. Under stated collision and conservative-validation assumptions, a finite cover of validated regions guarantees collision freedom throughout a certificate-backed continuous path, without sampled collision checks. The Shelf+IIWA warm profile completed all 40 tasks; median reusable-build time and within-batch completion timestamp were 0.171 and 0.077 s. Holding per-task time fixed, the ten-query build-plus-task projection was 69.6–79.7% below the one-query projection across three robots; this arithmetic result is not a measured multi-query or cross-revision speedup. At method-specific points selected on this six-condition catalog, each RBF profile completed its 100 instances. The median empirical-reference-normalized path-length ratios for RBF were numerically below the corresponding RRT-Connect medians in all six conditions and below the corresponding medians for probabilistic roadmaps (PRM) and Batch Informed Trees (BIT*) in five.

Index Terms—Motion planning, robot manipulators, multi-query planning, collision checking, configuration-space decomposition, link envelopes.

I. INTRODUCTION

Manipulators often solve sequences of related queries in semi-static workcells, where robot kinematics remain fixed but start-goal pairs and local clutter change. A reusable planner should amortize free-space construction across queries, isolate obstacle-independent kinematic evidence from revision-local validity, and retain conservative current-scene support for every path segment claimed to be collision-free.

Planning methods support different forms of reuse. Probabilistic roadmaps, experience graphs, and safety certificates reuse connectivity, prior paths, or clearance neighborhoods, whereas standard tree and batch planners construct query-local search structures [1]–[3]. Region-based methods—including certified decompositions, safe-box planners, and motion-planning Graphs of Convex Sets (GCS)—operate on scene-specific configuration-space sets and graphs [4]–[7]. This separation raises a central question: can a region graph be reconstructed for each scene revision while obstacle-independent kinematic evidence is reused through exact identity matching? Table I positions the proposed RapidBoxForest (RBF) framework and its Lifelong Envelope Cache Tree (LECT) relative to representative planning families.

Each LECT record and its corresponding primitive forest cell share a canonical dyadic address. Replay therefore uses exact-key lookup rather than geometric rematching, while validation and refinement operate on the same address space. Only obstacle-independent endpoint records persist across compatible revisions; RBF reconstructs link envelopes, box labels, and adjacency for the current scene. Coalesced scene nodes represent exact dyadic unions but do not serve as persistent cache keys.

a) *Problem statement*: Let $Q \subset \mathbb{R}^n$ be the joint-limit box of a fixed n -degree-of-freedom manipulator, and let $\{\mathcal{O}^{(r)}\}$ denote a sequence of revisions to the workspace obstacles. Endpoint-record identity κ_E specifies the robot endpoint model and canonical dyadic decomposition, whereas descriptor δ specifies the active-link geometry and validator settings. For fixed δ , $\mathcal{C}_{\text{free}}(\mathcal{O}^{(r)}) \subseteq Q$ denotes the modeled free configuration space; its dependence on δ is suppressed below. Each query provides $(q_s, q_g) \in Q^2$, and RBF constructs a finite graph of policy-accepted configuration-space boxes. Queries within one revision share this graph. Across revisions, an endpoint record is replayed only when both its joint region and κ_E match exactly; link envelopes, box labels, and connectivity are then recomputed. The objective is a resource-bounded volumetric planner whose collision-free claims follow solely from conservative current-scene witnesses.

A scene revision is *compatible* when it preserves κ_E and the canonical address semantics. The descriptor δ may change, but every δ -dependent link envelope, label, adjacency, and validation result must then be reconstructed for the current scene.

b) *Reuse and continuous-path semantics*: RBF separates persistent endpoint geometry from revision-local graph state. A returned path is certificate-backed only if a complete finite cover of conservative current-scene witnesses is retained; any unsupported transition leaves the returned path validation-required.

The paper makes three contributions:

- 1) *Cross-revision evidence interface*. We introduce an exact identity contract that maps every persistent endpoint record to one canonical dyadic forest cell. This contract permits reuse of obstacle-independent geometry while requiring all scene-dependent state to be reconstructed.
- 2) *Cache-aligned construction*. We develop a resource-bounded builder that replays or materializes endpoint records, refines unresolved canonical cells, and coalesces cells only through exact dyadic unions with retained provenance, thereby preserving primitive cache keys and the certified free set.

TABLE I
REPRESENTATIVE REUSE AND REVISION HANDLING; VARIANTS MAY DIFFER

Planning paradigm	Reused state	Explicit region representation	Revision handling	Implication for repeated queries
Roadmaps, experience graphs, and clearance caches	Vertices, edges, path fragments, or clearance neighborhoods	Method dependent	Revalidate, retrieve, or repair affected graph elements	Reuses connectivity or experience without maintaining a volumetric scene decomposition.
Tree-based and anytime sampling	Query-local tree or batch graph	Not in the standard formulation	Rebuild per independent query; checkpoints may persist within a run	Supports rapid discovery or refinement without a persistent volumetric representation.
Convex-region and safe-box planning	Scene-specific cells or boxes and adjacency	Yes	Revalidate, repair, or reconstruct scene-dependent regions after changes	Reuses an existing region graph within a fixed scene.
RBF	Revision-local typed box graph; obstacle-independent endpoint records across revisions	Yes	Replay exactly matched endpoint records; recompute link envelopes, box labels, and adjacency	Reuses kinematic evidence while deriving every collision-validity claim from the current scene.

3) *Continuous-path collision freedom.* We prove pointwise collision freedom for any returned path covered by finitely many conservatively validated axis-aligned boxes or oriented bounding boxes (OBBs), under the stated collision and numerical assumptions. Any unsupported tolerance-gap, segment, or repair transition leaves the returned path validation-required.

Sections III–V develop the method (Fig. 1); Q1–Q4 evaluate the warm-cache cost profile, planning trade-offs at the reported operating points, and endpoint- and link-envelope mechanisms.

II. RELATED WORK

Region-based guarantees vary by method. IRIS (Iterative Regional Inflation by Semidefinite Programming) constructs convex regions, configuration-space IRIS (C-IRIS) certifies polytopes [4], [8], and graphics-processing-unit (GPU) methods construct probabilistically collision-free convex sets [9], [10]. Visibility-graph clique covers reduce the number of sets [11]. Fast safe-box planning preprocesses a safe-region collection [5], whereas GCS optimizes trajectories over convex-region graphs [6], [12]. Implicit GCS methods (IxG/IxG*) and GCS* restrict search to query-relevant portions of an implicit region graph [13], [14]; multi-query GCS precomputes a cost-to-go surrogate for repeated queries on a fixed GCS [7]. These methods construct or search region graphs whose geometric sets and successor rules are scene dependent. RBF instead couples construction with revision handling in a dyadic region graph: its primitive cells index obstacle-independent endpoint envelopes, while all obstacle-dependent labels and overlaps are recomputed after a revision.

a) *Closest methods:* ReUse-based Probabilistic Roadmap (RU-PRM) reuses obstacle-local roadmaps by matching and transforming corresponding models across similar environments [15]; Ball Tree associates volumes with nodes of a single-query search tree [16]. Safety certificates cache local clearance [2]. Dynamic roadmaps, decomposed collision checking, and Simple Polyhedral Intersection Techniques (SPITE) reduce revalidation work for pre-existing sampled graphs; SPITE, in particular, associates configurations and edges with workspace swept-volume approximations and tests

those volumes against moved obstacles [17]–[19]. Experience graphs retain prior path or graph experience [3], whereas soft subdivision targets conservative cell classification [20]. These methods differ in the state retained across revisions. C-IRIS constructs scene-specific certified regions; GCS optimizes over a supplied scene-specific region graph. RU-PRM, dynamic roadmaps, and SPITE retain or update sampled graph elements; clearance caches retain local validity neighborhoods. Across revisions, RBF retains neither scene-validity labels nor connectivity. Instead, it reuses obstacle-independent endpoint envelopes indexed by canonical volumetric cells and reconstructs current-scene labels, overlaps, and corridor witnesses.

b) *Endpoint bounds:* Common tools for geometric proximity queries include the Gilbert–Johnson–Keerthi (GJK) algorithm [21] and the Flexible Collision Library (FCL) [22]. Interval methods provide conservative bounds for nonlinear robot kinematics [23], [24]. Articulated continuous collision detection (CCD) has used interval hierarchies, swept primitives, and Taylor models to bound motion [25], [26]. RBF adopts interval bounding within a cross-revision reuse contract: versioned, obstacle-independent endpoint envelopes may persist, whereas link envelopes and obstacle-separation tests are recomputed for the current scene.

c) *Graph reuse:* The probabilistic roadmap (PRM) amortizes sampled-roadmap construction across queries, whereas LazyPRM defers sampled-edge validation [1], [27]. Effort Informed Roadmaps (EIRM*), decomposed collision checking, and Repetition Roadmaps reuse prior edges, collision-test results, or paths [18], [28], [29]. Adaptive cell planners refine graphs of accepted cells [20], [30], [31]. RBF instead assigns canonical volumetric cells two roles: they index reusable, obstacle-independent endpoint records and form the primitives of a newly labeled overlap graph constructed under an explicit resource budget.

d) *Sampling and refinement:* RRT-Connect is a bidirectional single-query planner [32], and Batch Informed Trees (BIT*) is an informed, anytime, asymptotically optimal sampling-based planner [33]. The Open Motion Planning Library (OMPL) provides the benchmarking infrastructure [34]. RBF instead constructs reusable volumetric context before query-local repair or optimization.

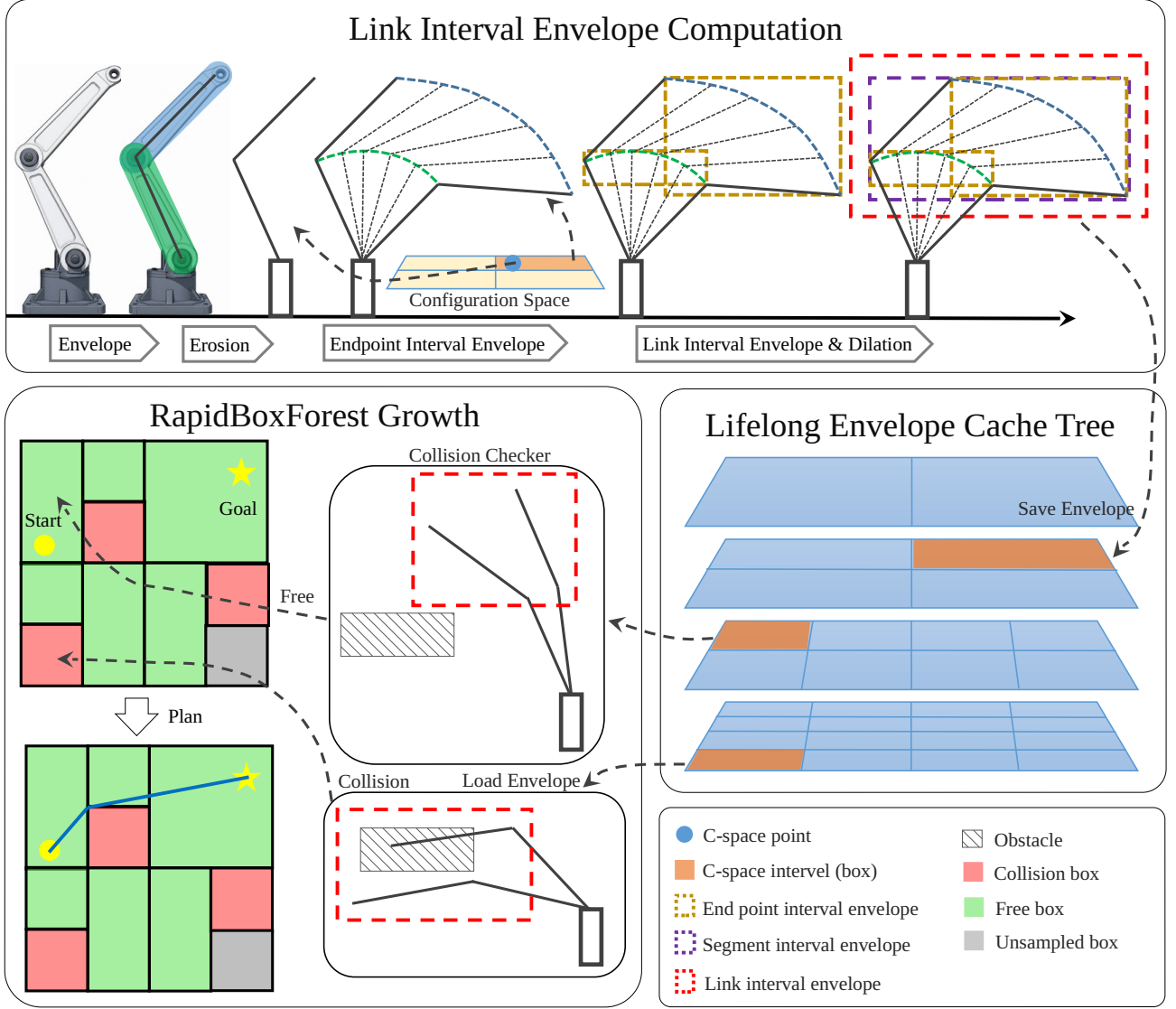


Fig. 1. RBF pipeline. Top: endpoint envelopes bound the zero-radius link centerline and are expanded by the link radius. Lower left: sample-guided growth produces accepted, rejected, and unsampled boxes. Lower right: LECT replays obstacle-independent endpoint records, while RBF recomputes link envelopes, obstacle labels, and adjacency for the current scene revision.

III. LINK INTERVAL ENVELOPES

The separation of endpoint and link representations enables cross-revision reuse. For a joint region $I \subseteq \mathcal{Q}$, $E_k(I)$ bounds the trajectory of endpoint k and is reusable under an identity match, whereas $B_\ell(I)$ is derived under the current descriptor and tested against the current scene:

$$I \mapsto \{E_k(I)\} \mapsto \{B_\ell(I)\}.$$

RBF uses axis-aligned boxes as global forest cells and OBB regions only as local finite-cover corridor witnesses. Only current-scene conservative validation lets either region contribute to a path certificate.

Assumption 1 (Collision model for certificates). *The certificate model covers collisions between active rounded links and a finite union $\mathcal{O}$ of closed workspace obstacles represented by axis-aligned bounding boxes (AABBs); boundary contact is collision, and $I \subseteq \mathcal{Q}$ enforces joint limits. A change to either*

the scene revision or descriptor δ invalidates every scene-dependent box and edge label. The semantic status CertFree denotes this conservative outcome; a runtime label without the required proof carries no free-space claim. Self-collision, attached bodies, grippers, and task constraints are outside this model and would require additional conservative validation.

a) Numerical contract: Soundness assumes outward-rounded endpoint and link envelopes, with all numerical margins included in δ . Here $\mathbb{B}_p(r)$ is the p -norm ball of radius r ; the AABB broadphase inflates by $\mathbb{B}_\infty(r_\ell)$, which contains the Euclidean link-radius ball. The soundness results below therefore apply to an implementation that enforces this numerical contract; all such results are conditional on it.

Definition 1 (Returned-path certificate status). Let ν_{scene} denote the current scene revision, let $\mathcal{O}(\nu_{\text{scene}})$ be its obstacle set, and fix descriptor δ and a continuous returned path $\gamma : [0, 1] \rightarrow \mathcal{Q}$. A retained finite-cover witness $W = \{(J_j, R_j)\}_{j=1}^m$ consists of closed parameter windows $J_j \subseteq [0, 1]$ and regions $R_j \subseteq \mathcal{Q}$ satisfying $[0, 1] = \bigcup_{j=1}^m J_j$ and $\gamma(J_j) \subseteq R_j$. Each retained R_j is either (i) an axis-aligned box for which the current validator returns CertFree, (ii) an axis-aligned box that is the exact union of same-revision, descriptor-matched CertFree boxes with a retained verified-tiling record, or (iii) a complete C-space OBB region for which the affine-zonotope validator returns CertFree under the same scene revision and descriptor. The path is certificate-backed exactly when such a retained W exists; otherwise it is validation-required.

A. Geometric Foundation

For a body-frame shape C_0 , let $T(q)x = R(q)x + t(q)$ map point x to workspace, with rotation $R(q)$ and translation $t(q)$, and let $\mathcal{S}_I(C_0)$ denote its sweep over $q \in I$ and $\oplus$ the Minkowski sum. For a capsule $C = S \oplus \mathbb{B}_2(r)$ with center segment S , rigid motion gives

$$\mathcal{S}_I(C) = \mathcal{S}_I(S) \oplus \mathbb{B}_2(r).$$

Thus a rounded link can be bounded by first bounding the zero-radius center segment and then lifting by the link radius. Let $\Gamma_a(I)$ and $\Gamma_b(I)$ denote the exact workspace trajectories of a capsule link's proximal and distal endpoints over I . RBF stores conservative endpoint bounds in reusable records keyed by kinematics; the following proposition converts them into whole-link envelopes.

Proposition 1 (Endpoint envelopes induce link interval envelopes). Consider a rounded link with body-frame endpoints a, b , $L = [a, b] \oplus \mathbb{B}_2(r)$, and any joint region $R \subseteq \mathcal{Q}$. If endpoint envelopes $E_a(R)$ and $E_b(R)$ satisfy $\Gamma_a(R) \subseteq E_a(R)$ and $\Gamma_b(R) \subseteq E_b(R)$, then

$$\mathcal{S}_R(L) \subseteq \text{conv}(E_a(R) \cup E_b(R)) \oplus \mathbb{B}_2(r).$$

Consequently, any conservative outer representation of the right-hand side is a valid link envelope for the whole joint region R .

Proof. For each $q \in R$, the moved center segment is the convex hull of its endpoints. Hence the union of these segments lies in $\text{conv}(\Gamma_a(R) \cup \Gamma_b(R)) \subseteq \text{conv}(E_a(R) \cup E_b(R))$. Adding the fixed-radius ball then yields the capsule sweep and the stated containment. $\square$

Corollary 1 (Conservative configuration-space box validation). Let $I \subseteq \mathcal{Q}$ be an axis-aligned configuration-space box and let $\{B_\ell(I)\}$ be conservative link interval envelopes for all active rounded links. If $B_\ell(I) \cap \mathcal{O} = \emptyset$ for every link ℓ , then every configuration $q \in I$ is collision-free with respect to the collision model in Assumption 1.

Proof. By conservatism, each $B_\ell(I)$ contains the complete workspace sweep of link ℓ over I . Its disjointness from $\mathcal{O}$ therefore excludes the modeled workspace-obstacle collision for every active link and every $q \in I$ under Assumption 1. $\square$

B. Endpoint Interval Envelopes

The cached primitive is an interval AABB (iAABB) enclosing the endpoint trajectory $\Gamma_k(I)$ and shared by all link representations. Affine-arithmetic interval forward kinematics (IFK-AA) uses one Denavit–Hartenberg interval pass. Hierarchical interval forward kinematics (HIFK) bisects I to depth d and hulls the child boxes, denoted HIFK- d . Within this certificate model, Critical-sample bounds are filter-only and ineligible for CertFree; Q4 also reports Analytical and Monte Carlo empirical references. Fig. 1 summarizes the conversion.

C. C-space OBB Endpoint Envelopes

Axis-aligned boxes remain the global forest primitive; an OBB is local edge evidence for a bridge or repair. Let q_c be its center, let the diagonal matrix $S \succ 0$ scale the joints, let $U \in \mathbb{R}^{n \times m}$, $1 \leq m \leq n$, have orthonormal columns in scaled coordinates, and let $h \in \mathbb{R}_{\geq 0}^m$ contain the half-widths. With $A = S^{-1}U \text{diag}(h)$ and $\Xi = \{\xi \in \mathbb{R}^m : -1 \leq \xi_i \leq 1\}$, the joint-space region is

$$R_{\text{obb}} = \{q_c + A\xi \mid \xi \in \Xi\}.$$

Here “OBB” denotes this affine image; it may be lower dimensional when $m < n$, and the full-dimensional case has $m = n$. For R_{obb} , the correlated validator bounds each endpoint trajectory using the outward-rounded affine-zonotope enclosure $E_k(R_{\text{obb}})$ specified below. Certificate eligibility requires $R_{\text{obb}} \subseteq \mathcal{Q}$, verified containment of the represented path window, and conservative validation of the complete region; the fitted orientation alone provides no safety status.

a) *Finite path cover:* A piecewise-linear candidate path γ has an OBB witness only if finitely many closed windows J_τ cover $[0, 1]$, with $\gamma(J_\tau) \subseteq R_\tau \subseteq \mathcal{Q}$, and every complete R_τ passes the correlated endpoint/link test. Any uncovered window leaves the path validation-required; sampled fallbacks may guide search but cannot establish certificate status.

Corollary 2 (OBB-region and finite-cover path soundness). Fix a scene revision ν_{scene} and descriptor δ , and write $\mathcal{O} = \mathcal{O}(\nu_{\text{scene}})$. Let $\{R_\tau\}_{\tau=1}^M$ be a finite family of joint-space OBB regions contained in $\mathcal{Q}$. Suppose that, for every τ , conservative affine-zonotope endpoint enclosures induce conservative envelopes for all active links over the complete region R_τ , and that those link envelopes pass conservative separation tests against the current-scene obstacles under the same revision and descriptor. Then $R_\tau \subseteq \mathcal{C}_{\text{free}}(\mathcal{O})$ for every τ . Consequently, if closed windows J_τ cover $[0, 1]$ and a continuous path γ satisfies $\gamma(J_\tau) \subseteq R_\tau$ for every τ , then $\gamma([0, 1]) \subseteq \mathcal{C}_{\text{free}}(\mathcal{O})$ under Assumption 1.

Proof. For each τ , Proposition 1 and current-scene separation give $R_\tau \subseteq \mathcal{C}_{\text{free}}(\mathcal{O})$. Under the additional cover hypothesis, every $t \in [0, 1]$ belongs to some J_τ , and hence $\gamma(t) \in R_\tau \subseteq \mathcal{C}_{\text{free}}(\mathcal{O})$. The path claim follows. $\square$

b) *Correlated endpoint envelope:* For $q = q_c + A\xi$, outward affine propagation gives

$$E_k(R_{\text{obb}}) = c_k + \sum_{i=1}^p g_{k,i} \epsilon_i \oplus [-\rho_k, \rho_k].$$

Here p is the number of affine generators, $c_k, g_{k,i} \in \mathbb{R}^3$, $\epsilon_i \in [-1, 1]$, and $\rho_k \in \mathbb{R}_{\geq 0}^3$; $[-\rho_k, \rho_k]$ is componentwise and ρ_k outwardly encloses the entire nonlinear remainder. A sampled remainder is filter-only. For workspace direction $d \in \mathbb{R}^3$, the support of E_k is

$$h_{E_k}(d) = d^\top c_k + \sum_{i=1}^p |d^\top g_{k,i}| + \sum_{\mu=1}^3 |d_\mu| \rho_{k,\mu}.$$

The rounded-link support bound is therefore

$$\bar{h}_\ell(d) = \max\{h_{E_a}(d), h_{E_b}(d)\} + r_\ell \|d\|_2.$$

Coordinate directions provide the AABB broadphase; retained supports provide the SupportHull/GJK test. OBB records remain scene-local typed-edge evidence and do not become persistent LECT records or global forest nodes.

D. Link-Envelope Representations

Endpoint records can be exported to multiple link representations without changing the upstream kinematic source. For a serial-chain capsule link ℓ whose proximal and distal endpoint records are $E_{\ell-1}(I)$ and $E_\ell(I)$, $B_\ell(I) = B_\ell^\circ(I) \oplus \mathcal{R}(r_\ell)$, with $B_\ell^\circ(I)$ bounding the centerline sweep and $\mathcal{R}(r_\ell) \supseteq \mathbb{B}_2(r_\ell)$ denoting a conservative radius-lift set. AABB uses $\mathbb{B}_\infty(r_\ell)$, while SupportHull uses the Euclidean-ball support.

a) AABB envelope: Each AABB envelope encloses the two endpoint envelopes in a single box and expands it by the link radius:

$$B_\ell(I) = \text{bbox}(E_{\ell-1}(I) \cup E_\ell(I)) \oplus \mathbb{B}_\infty(r_\ell).$$

Here $\text{bbox}(\cdot)$ denotes the workspace AABB hull.

b) SupportHull envelope: SupportHull stores support mappings for both endpoint enclosures and the link radius. The default implementation uses endpoint AABBs; correlated OBB corridors use the zonotope-plus-box records above. A bounded-work GJK narrow phase uses outward obstacle inflation. If the iteration limit is reached without a decision, the method returns Unknown rather than a free label.

Fig. 2 illustrates the resulting build, query, and update cycle on a two-link schematic.

IV. LIFELONG ENVELOPE CACHE TREE

LECT supplies the cross-revision half of the reuse contract by indexing obstacle-independent endpoint evidence with the dyadic joint-space boxes used in forest construction. RBF derives link envelopes under the current descriptor and refreshes every scene-dependent label; graph connectivity and query outcomes are never reused across scene revisions.

A. Cache Key and Split Policy

LECT is a dynamically grown binary k -d tree over $\mathcal{Q}$; each node stores a joint interval, split metadata, and geometry records.

a) Cache identity: Replay requires exact matches of endpoint-record identity κ_E and dyadic address. The identity binds robot kinematics, endpoint representation, root canonicalization, split policy, schema, and builder version, whereas δ binds link geometry and validation settings; only endpoint records persist. LECT and the forest follow one deterministic midpoint split schedule, aligning records with primitive cells without geometric rematching.

Formally, let κ_E pass the endpoint-record gate above, fixing a common root box, canonicalization, and split schedule s_{κ_E} . For a root-relative child address $\alpha = (c_1, \dots, c_d)$, $c_i \in \{0, 1\}$, the shared decoder maintains

$$I_{\kappa_E}^{\text{LECT}}(\alpha) = I_{\kappa_E}^{\text{forest}}(\alpha) = I_{\kappa_E}(\alpha).$$

For endpoint k , each stored record $(\kappa_E, \alpha, k, E_k)$ satisfies the record invariant

$$\Gamma_k(I_{\kappa_E}(\alpha)) \subseteq E_k.$$

Certificate-eligible coalesced nodes must retain an exact-union record and the matched revision–descriptor label; they never become persistent cache keys. Constituent boxes may leave the active graph only while that record remains. This identity invariant, rather than geometric lookup, provides the interface between cache replay and forest refinement.

b) Sparse interval staging: Planning may stage a canonical dyadic interval at address α without materializing all of its ancestors. If its endpoint record is absent, the record is materialized under (κ_E, α) before current-scene validation.

B. Runtime Use and Validation Semantics

For a joint interval, LECT replays or materializes endpoint evidence, derives link envelopes under δ , and applies the current-scene validator. When child endpoint records are hulled to form a parent endpoint enclosure, link envelopes are re-derived for the parent joint cell and validated against the current scene; child labels are not inherited. Missing or inconclusive evidence yields Unknown. Cache hits may change enclosure tightness and hence the acceptance outcome, but not the meaning or validation obligations of any returned status.

The cache-identity gate above controls replay eligibility; soundness additionally requires decoded containment and fresh conservative scene validation.

Proposition 2 (Sound replay of matched endpoint enclosures). *Let $I = I_{\kappa_E}(\alpha) \subseteq \mathcal{Q}$, and fix the current scene revision ν_{scene} and descriptor δ . Suppose each required endpoint record matches the current κ_E and satisfies the record invariant above at address α . If the link envelopes derived under δ are conservative and all pass conservative separation tests against $\mathcal{O}(\nu_{\text{scene}})$, then $I \subseteq \mathcal{C}_{\text{free}}(\mathcal{O}(\nu_{\text{scene}}))$ under Assumption 1.*

Proof. The matched record invariant supplies every endpoint inclusion over I . Proposition 1 therefore places each rounded-link sweep inside its derived conservative envelope. Current-scene separation and Corollary 1 give the stated inclusion. $\square$

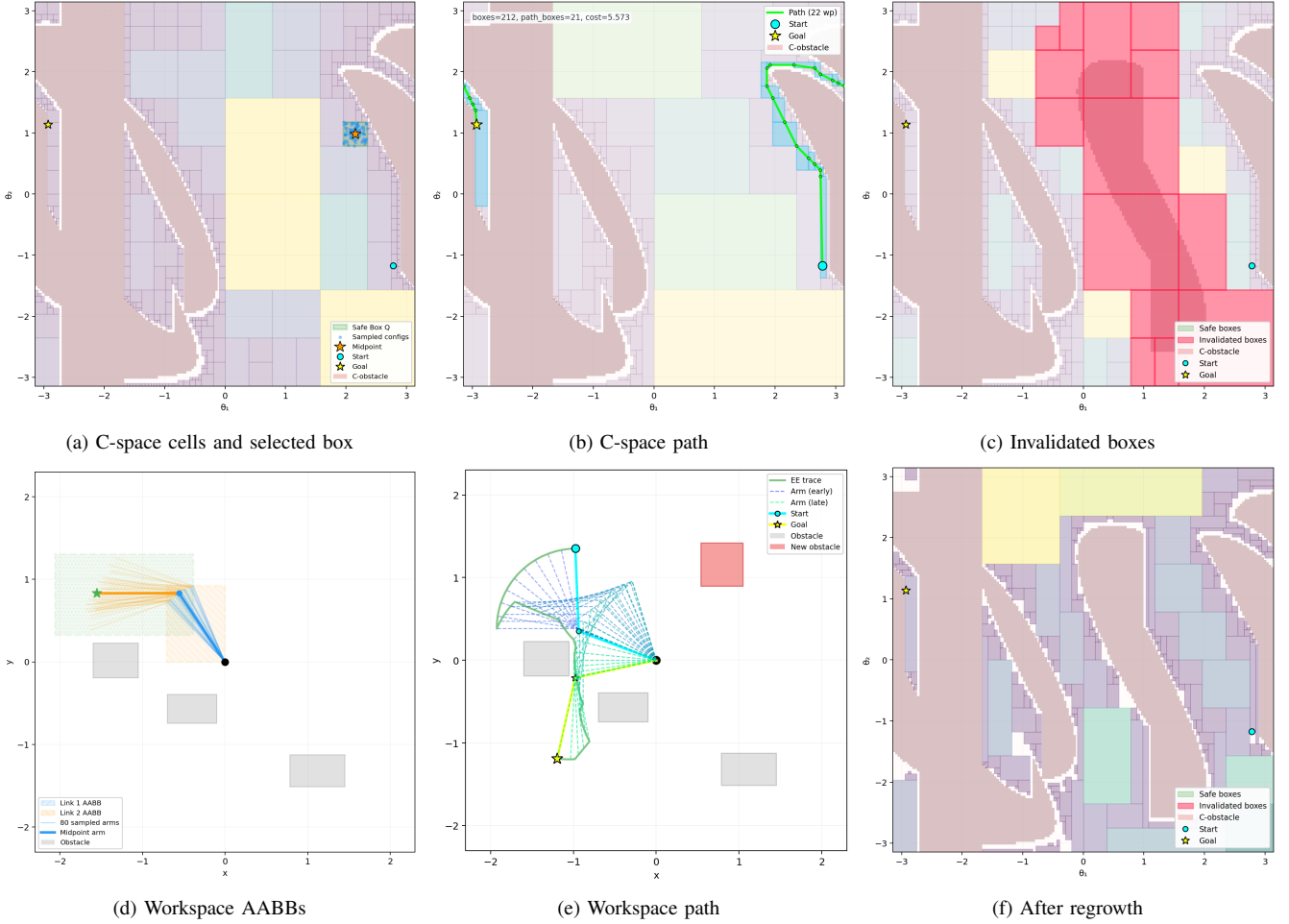


Fig. 2. Two-link RBF build-query-update schematic. Top: (a) selected C-space box, (b) route, and (c) invalidation; bottom: (d) workspace link AABBs, (e) mapped route, and (f) regrowth. Embedded counts and costs are illustrative; θ_0, θ_1 are joint coordinates and x, y are workspace coordinates; end-effector (EE) and waypoint (wp) are abbreviated in panels (b) and (e).

C. Storage and Revision Semantics

Serialization retains compatible endpoint geometry and split metadata only; link envelopes, labels, adjacency, and query outcomes are reconstructed per revision.

V. RAPIDBOXFOREST CONSTRUCTION

Fig. 2 summarizes the planning layer. Shared dyadic cell identities let replay, batch validation, and refinement address the same joint boxes. RBF turns envelope validation and LECT replay into a query-reusable, scene-specific structure comprising policy-accepted C-space boxes and typed corridor adjacencies. Queries then use box membership and graph search. A box can support W only through current-scene conservative validation or a retained same-revision exact-union witness.

A box is an axis-aligned joint interval $B = [l^{(1)}, u^{(1)}] \times \dots \times [l^{(n)}, u^{(n)}] \subseteq Q$. A *policy-accepted box* has passed the configured scene-validation policy. Under the stated geometric and numerical assumptions, conservative link-envelope disjointness makes it eligible for CertFree; filter-only acceptance yields a provisional region that cannot contribute to W . Rejected or unresolved cells may be retained for refinement but are excluded from the forest.

A. Budgeted Construction Contract

Construction returns a resource-bounded reusable region graph. Given the joint-limit box Q and obstacle set $\mathcal{O}$, the scene policy $V_{\mathcal{O}}(B; \delta)$ evaluates active-link envelopes for B under descriptor δ and has range

$$V_{\mathcal{O}}(B; \delta) \in \{\text{CertFree}, \text{ProvisionalFree}, \text{Occupied}, \text{Unknown}\}.$$

By definition of the conservative branch,

$$V_{\mathcal{O}}(B; \delta) = \text{CertFree} \implies B \subseteq \mathcal{C}_{\text{free}}(\mathcal{O}).$$

ProvisionalFree is filter-only, Occupied denotes a policy rejection rather than a certificate of collision, and Unknown is inconclusive. The accepted region set of N boxes is

$$\mathcal{F} = \{(B_i, \sigma_i) : i = 1, \dots, N\}, \\ B_i \subseteq Q, \quad \sigma_i \in \{\text{CertFree}, \text{ProvisionalFree}\}.$$

Algorithm 1 states the certificate-producing branch and therefore commits only boxes with semantic CertFree status. Filter-only profiles may retain ProvisionalFree boxes for search, but those boxes cannot contribute to W without conservative revalidation. For a same-revision family $\mathcal{G}$ with

matching descriptor, let $H = \text{bbox}(\bigcup_{B \in \mathcal{G}} B)$; the hull inherits this label exactly when

$$\text{ExactUnion}(H, \mathcal{G}) \iff H = \bigcup_{B \in \mathcal{G}} B$$

holds. Let ε_c denote the connection tolerance, with $\varepsilon_c \geq 0$; it does not enter this predicate. Exact union is tested by dyadic integer-range coverage, not floating-point volume equality, and is enabled in Q1 and at the Q2 RBF operating point but disabled in Q3. Let $\mathcal{S} \subseteq \mathcal{Q}$ be a finite anchor set fixed before queries; it affects admission priority, not $V_{\mathcal{O}}$ or certificate status. The reusable scene state $(\mathcal{F}, G_{\cap})$ is certificate-eligible only with recorded revision, descriptor, and exact-union provenance; query-specific typed edges are added only during corridor construction.

For two boxes

$$B_i = \prod_{d=1}^n [l_i^{(d)}, u_i^{(d)}], \quad B_j = \prod_{d=1}^n [l_j^{(d)}, u_j^{(d)}],$$

RBF treats the boxes as candidate-adjacent when their closed intervals in every coordinate overlap or have a gap no larger than ε_c :

$$\max\{l_i^{(d)}, l_j^{(d)}\} \leq \min\{u_i^{(d)}, u_j^{(d)}\} + \varepsilon_c, \quad d = 1, \dots, n.$$

Only exact overlap or contact directly yields a contained box corridor. Tolerance-gap, segment, and repair edges are search aids; they support a certificate only after conversion to a conservatively validated box corridor or finite-cover OBB corridor.

B. Build Pipeline

Let $\mathcal{T}_L$ denote the LECT tree, and let $G_{\cap} = (\mathcal{F}, E_{\cap})$ be the reusable exact-overlap/contact subgraph. Algorithm 1 batch-validates a LECT-aligned frontier, coalesces exact unions, and refines unresolved regions under an anchor-prioritized budget. Its $\widehat{V}_{\mathcal{O}}(I; \kappa_E, \delta)$ operation obtains or materializes matched endpoint evidence, derives and validates current link envelopes, and maps missing or inconclusive evidence to Unknown. It returns $\mathcal{F}$, $G_{\cap}$, and deferred frontier state; attachments and typed transitions are query-local.

C. Corridor Query and Certificate

The typed graph is a query-local overlay on $G_{\cap}$: $G = (\mathcal{F}, E_{\cap} \cup E_{\varepsilon} \cup E_s \cup E_{\text{obb}} \cup E_{\text{rep}})$, comprising overlap/contact, tolerance gap, candidate-segment, finite-cover OBB-corridor, and repair edges. Start/goal attachments remain separate. Edge types record candidate origin, whereas Definition 1 alone determines certificate status.

For a returned route, an overlap/contact transition contributes its incident validated boxes and contained gateway; an OBB transition contributes its retained finite cover. Other edges and endpoint attachments contribute no witness until converted to a conservatively validated box corridor or finite-cover OBB corridor. Concatenating the parameter windows constructs W ; certificate status requires full coverage without an unsupported transition.

Algorithm 1 Zero-gap coalescing and budgeted refinement.

Input: anchors $\mathcal{S}$; LECT $\mathcal{T}_L$; $\kappa_E, \mathcal{O}, \nu_{\text{scene}}, V_{\mathcal{O}}, \delta$; depths $0 \leq D_0 \leq D_s$; admission cap $B_{\max} \in \mathbb{Z}_{\geq 1}$; time, depth, and retention caps

Output: $\mathcal{F}, G_{\cap}, \mathcal{C}$

- 1: form the identity-deduplicated $\mathcal{S}$ - $\mathcal{T}_L$ frontier at depths $D_0:D_s$; batch-evaluate $s_I \leftarrow \widehat{V}_{\mathcal{O}}(I; \kappa_E, \delta)$
- 2: for each same-revision, descriptor-matched family $\mathcal{G}$ of CertFree cells, set $H \leftarrow \text{bbox}(\bigcup_{B \in \mathcal{G}} B)$
- 3: coalesce only if dyadic integer ranges establish $\text{ExactUnion}(H, \mathcal{G})$
- 4: insert $(H, \text{CertFree})$ with its exact-union record under $(\nu_{\text{scene}}, \delta)$; remove constituents only from the active set
- 5: insert unmerged CertFree leaves into $\mathcal{F}$; build $G_{\cap}$
- 6: initialize $\mathcal{P}$ with maximal frontier cells, deduplicated by identity and prioritized by anchors; $\mathcal{C} \leftarrow \emptyset$; $b \leftarrow 0$
- 7: **while** $\mathcal{P}$ is nonempty, $b < B_{\max}$, and time/depth caps remain **do**
- 8: pop and batch-evaluate $\mathcal{A} \subseteq \mathcal{P}$ with $|\mathcal{A}| \leq B_{\max} - b$; $b \leftarrow b + |\mathcal{A}|$
- 9: for each $I \in \mathcal{A}$, commit it and update $G_{\cap}$ if $s_I = \text{CertFree}$; otherwise enqueue unseen LECT-aligned children or defer unsplitable I to $\mathcal{C}$
- 10: **end while**
- 11: $\mathcal{C} \leftarrow$ retention-capped $(\mathcal{C} \cup \mathcal{P})$; record omissions
- 12: **return** $\mathcal{F}, G_{\cap}, \mathcal{C}$

A best-first traversal orders candidate routes heuristically. For overlap routes, gateways lie in consecutive box intersections, and convexity keeps their connecting segments within witnessed boxes.

Lemma 1 (Box-corridor certificate). *For scene revision ν_{scene} and descriptor δ , let $\mathcal{F}_{\text{cert}}$ contain boxes labeled CertFree against $\mathcal{O}(\nu_{\text{scene}})$, each directly validated or retained as a zero-gap exact union of such boxes under the same $(\nu_{\text{scene}}, \delta)$ pair. Set $R_{\text{cert}} = \bigcup_{B \in \mathcal{F}_{\text{cert}}} B$. If a continuous path $\gamma : [0, 1] \rightarrow \mathcal{Q}$ lies in R_{cert} , then $\gamma([0, 1]) \subseteq \mathcal{C}_{\text{free}}(\mathcal{O}(\nu_{\text{scene}}))$ under Assumption 1.*

Proof. Directly validated boxes lie in $\mathcal{C}_{\text{free}}$ by Corollary 1; each coalesced box is an exact union of such boxes under the same $(\nu_{\text{scene}}, \delta)$. Hence $R_{\text{cert}} \subseteq \mathcal{C}_{\text{free}}$, and the path inclusion follows. $\square$

Corollary 3 (Route construction from an overlap corridor). *Let $B_1, \dots, B_m \in \mathcal{F}_{\text{cert}}$, with $q_s \in B_1$, $q_g \in B_m$, and gateways $g_i \in B_i \cap B_{i+1}$. The polyline $(q_s, g_1, \dots, g_{m-1}, q_g)$ lies in R_{cert} and is collision-free under Assumption 1. Replacing a route segment by a continuous segment whose complete image is covered by a retained finite cover of validated OBB regions preserves the conclusion. Any transition lacking a retained validated-box or OBB-cover witness makes the returned route validation-required.*

Proof. Let $g_0 = q_s$ and $g_m = q_g$. For each $i = 1, \dots, m$, both endpoints of $[g_{i-1}, g_i]$ lie in the convex box B_i , so the complete polyline lies in R_{cert} and Lemma 1 applies. A validated OBB replacement follows from Corollary 2. $\square$

Theorem 1 (Continuous collision freedom). *Under Assumption 1 and the numerical contract above, fix a scene revision ν_{scene} and descriptor δ . For any continuous returned path $\gamma : [0, 1] \rightarrow \mathcal{Q}$ for which a conforming implementation retains a witness W satisfying Definition 1,*

$$\forall t \in [0, 1], \quad \gamma(t) \in \mathcal{C}_{\text{free}}(\mathcal{O}(\nu_{\text{scene}})).$$

TABLE II
EVALUATION QUESTIONS, EVIDENCE, AND REUSE MECHANISMS

Question	Evaluation objective	Main evidence	Reuse mechanism
Q1 Shelf profile	Characterize the registered warm-cache profile and control costs.	One no-cache contrast and three registered cache-backed controls over eight seeds.	Endpoint replay and scene-local rebuild.
Q2 Shelf comparison	Contextualize selector-conditioned path length and within-method trade-offs.	Five methods over eight seeds.	Within-scene graph reuse over five queries.
Q3 Random scenes	Characterize selected in-sample profiles across robot models.	Six robot-difficulty conditions, 100 queries each.	Within-scene graph reuse over ten queries.
Q4 Mechanisms	Isolate endpoint- and link-envelope trade-offs.	1000 shared boxes per width at six widths.	Endpoint construction and link-envelope validation.
Cache update	Record scene-cache maintenance primitives.	Historical insertion, removal, and warm rebuild.	Additional timing outside Q1–Q4.

TABLE III
SHELF+IIWA RBF WARM-D23 BASELINE AND REGISTERED PROFILE CONTROLS

Case	Build	Batch timestamp	$L_q/L_{\text{ref},q}$
RBF baseline	0.171 [0.166, 0.174]	0.077 [0.072, 0.085]	1.16 [1.09, 1.40]
Critical-sample	0.279 [0.275, 0.283]	0.087 [0.078, 0.095]	1.16 [1.10, 1.38]
Link-AABB	0.167 [0.164, 0.171]	0.083 [0.077, 0.086]	1.21 [1.09, 1.50]
HIFK-5/no cache	2.770 [2.644, 3.003]	0.181 [0.139, 0.207]	1.16 [1.10, 1.44]
SupportHull-only	0.193 [0.188, 0.200]	0.087 [0.082, 0.095]	1.16 [1.09, 1.40]

Notes: Seconds; medians [25%, 75%]; b100, 8 seeds. Batch timestamp: 40 within-batch completion times, not single-query latency. Baseline: IFK-AA+d23+Link-AABB+SupportHull/GJK. Registered caches: none (HIFK-5); d23 (Critical-sample); IFK-AA+d23 (Link-AABB and SupportHull-only). Critical-sample is non-certifying; its outputs remain validation-required.

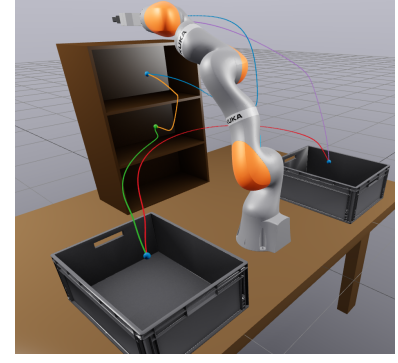


Fig. 3. Canonical Marcucci shelf scene [5] with a KUKA LBR iiwa manipulator (Shelf+IIWA) and the five fixed anchors.

Proof. For any $t \in [0, 1]$, the finite cover in W supplies a retained region R_j with $\gamma(t) \in R_j$. A directly validated box lies in $\mathcal{C}_{\text{free}}$ by Corollary 1, and an exact-union box inherits this inclusion from its retained verified-tiling record. Each validated OBB region lies in $\mathcal{C}_{\text{free}}$ by Corollary 2. Hence every $\gamma(t)$ is collision-free under the fixed revision and descriptor. This pointwise argument is independent of waypoint spacing or sampled checks. $\square$

Corollary 4 (Cross-revision path soundness). *After an obstacle revision from ν_{scene} to ν'_{scene} that preserves κ_E , fix the current descriptor δ . Suppose every replayed endpoint record matches κ_E and satisfies the record invariant, and every witness region is either directly revalidated against $\mathcal{O}(\nu'_{\text{scene}})$ under δ or newly constructed by a verified exact-union operation on ν'_{scene} -matched, δ -matched CertFree boxes. The certificate-backed returned path is then collision-free for every $t \in [0, 1]$ in the revised scene.*

Proof. Matched replay followed by fresh current-scene validation makes each primitive box sound by Proposition 2. Directly revalidated OBB regions are sound by Corollary 2, and a newly verified exact union preserves the inclusion of its current-revision constituents. Thus the retained witness satisfies Definition 1 for $(\nu'_{\text{scene}}, \delta)$, and Theorem 1 applies to its complete cover. $\square$

VI. EXPERIMENTS

Theorem 1 establishes continuous-path collision freedom from conservative finite-region coverage; the evaluation does not attempt to re-establish this guarantee by sampling. Instead, it addresses four questions about operating cost and planning behavior. Q1 characterizes a fixed warm-cache Shelf+KUKA LBR iiwa (Shelf+IIWA) profile and compound profile controls. Q2 provides path-length context against roadmap, sampling, and convex-region baselines. Q3 characterizes selected in-sample profiles on three robots, and Q4 isolates endpoint- and link-envelope trade-offs. Fig. 3 introduces the fixed Shelf scene. The Q2 baseline couples iterative regional inflation with Graphs of Convex Sets (IRIS/GCS). Table II maps each empirical question to its evidence and reuse mechanism.

Protocol and metrics. Experiments were run on Ubuntu 22.04 with an Intel Core i5-12600KF. OMPL planners were single-threaded; timings imported from archived runs without host-load metadata are treated as descriptive.

Build denotes post-initialization reusable construction time. For RBF, this metric includes evidence access or materialization, scene labeling, graph and anchor construction, and shortcut generation, but excludes snapshot creation/opening; PRM endpoints are preloaded. Online/q denotes per-query work. Timings exclude simplification. Successful runtime completion and the subsequent sampled check are empirical gates. Each saved polyline was subsequently densified to at most 0.01 rad between samples and collision-checked; this check was untimed. A failed gate excludes the record and can

TABLE IV
SHELF+IIWA COMPARISON AT METHOD-SPECIFIC, IN-SAMPLE OPERATING POINTS

Query	RBF b100 Build 0.171 s		IRIS/GCS+post Build 385.950 s		PRM Build 20.040 s		RRT-Connect No reusable build		BIT* No reusable build	
	Batch timestamp (s)	$L_q/L_{\text{ref},q}$	T (s)	$L_q/L_{\text{ref},q}$	T (s)	$L_q/L_{\text{ref},q}$	T (s)	$L_q/L_{\text{ref},q}$	T (s)	$L_q/L_{\text{ref},q}$
AS→TS	0.065	1.40	0.540	1.15	0.198	1.61	0.034	2.27	4.004	1.06
	[0.064, 0.070]	[1.34, 1.65]	[0.390, 0.661]	[1.11, 1.18]	[0.147, 0.248]	[1.42, 1.73]	[0.024, 0.052]	[2.05, 2.72]	[4.002, 4.004]	[1.02, 1.08]
TS→CS	0.085	1.13	0.346	1.22	0.173	2.99	0.054	3.98	10.138	1.09
	[0.082, 0.089]	[1.11, 1.14]	[0.296, 0.396]	[1.11, 1.28]	[0.118, 0.312]	[2.11, 3.22]	[0.021, 0.089]	[2.80, 5.09]	[10.004, 10.386]	[1.04, 1.11]
CS→LB	0.073	1.50	0.411	1.03	0.136	1.53	0.032	2.24	7.504	1.05
	[0.070, 0.077]	[1.35, 1.85]	[0.314, 0.518]	[1.02, 1.05]	[0.120, 0.195]	[1.42, 2.13]	[0.027, 0.070]	[1.79, 2.56]	[7.504, 7.725]	[1.04, 1.08]
LB→RB	0.078	1.10	0.761	1.02	0.164	1.20	0.001	1.31	1.654	1.03
	[0.075, 0.083]	[1.08, 1.24]	[0.685, 0.879]	[1.01, 1.06]	[0.091, 0.399]	[1.11, 1.29]	[0.001, 0.002]	[1.29, 1.34]	[1.654, 1.654]	[1.03, 1.04]
RB→AS	0.081	1.05	0.537	1.00	0.252	1.24	0.001	1.08	0.204	1.02
	[0.077, 0.086]	[1.04, 1.05]	[0.418, 0.609]	[1.00, 1.00]	[0.096, 0.295]	[1.18, 1.25]	[0.001, 0.001]	[1.04, 1.10]	[0.204, 0.204]	[1.01, 1.03]

Notes: Seconds; medians [25%, 75%], eight seeds. RBF Batch timestamp: 40 within-batch completions from eight five-query runs; other T values are per query. Times are method-local; postprocessing is excluded.

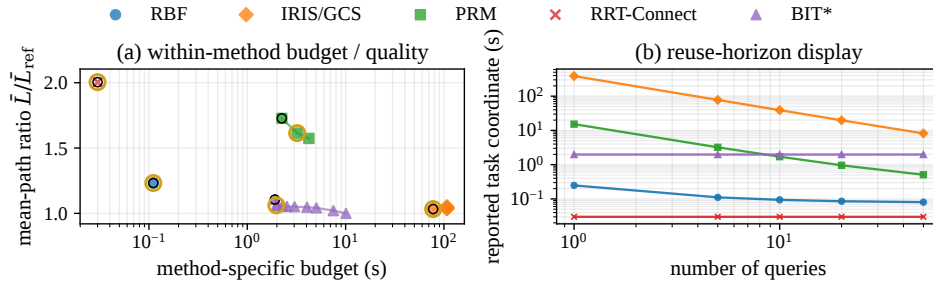


Fig. 4. Shelf+IIWA trade-offs over eight seeds. Left: mean path-length ratio $\bar{L}/\bar{L}_{\text{ref}}$. Black rings mark the first operating point that completes all 40 tasks; gold rings mark the registered RBF profile and each comparison method's fastest full-success point within 8% of its best mean path-length ratio. Right: method-local accounting projections over K ($K > 5$ extrapolated), not end-to-end latency; the RBF projection excludes the one-time d23 prewarm (1008.6 s; 12.74 GiB serialized cache). Table IV gives the Q2 selections.

change reported completion, but no sampled gate establishes W or supports Theorem 1. The results API and archives omit a complete witness W , so no per-path certificate-coverage rate is reported; experiments measure cost, completion, and path quality. The theorem applies only to a final path with retained W ; repair or postprocessing requires reconstructing W .

In Q1, Batch timestamp denotes a query's within-run completion time in a jointly scheduled five-query run (40 observations nested within eight runs). Q2 reports eight seeds per query, and each Q3 condition contains 100 queries nested within ten scenes. Q3 uses

$$\text{Task/q} = \left(T_{\text{adapt}}^{\text{wall}} + \sum_{i=1}^{N_q} T_{\text{query},i}^{\text{internal}} \right) / N_q.$$

Here, N_q is the batch size; the two terms are scene-adaptation wall time and internal query time. For accounting only, the reuse projections use unrounded metric values and add Build/ K to a fixed Task/q value; Shelf $K > 5$ and random-scene $K > 10$ are extrapolated. Tables report descriptive medians and interquartile ranges, not confidence intervals. Let L denote joint-space path length for a saved scene-query instance λ . We report $L/L_{\text{ref},\lambda}$, where $L_{\text{ref},\lambda}$ is the empirical minimum computed from the same data and pooled over the fixed methods and checkpoints before selection; it is neither a known optimum nor a held-out reference. Shelf uses the corresponding per-query reference. In Fig. 4, $\bar{L}$ and

$\bar{L}_{\text{ref}}$ denote the displayed mean path length and its empirical reference minimum, respectively.

Profile inheritance and cache accounting. The Q1 baseline and Q2 RBF operating point use a read-only depth-23 LECT and a 100-box admission cap (d23/b100). Q1 controls follow registered, profile-specific cache assignments rather than a common cache treatment: HIFK-5 is the no-cache contrast; Critical-sample replays its matching d23 cache; and Link-AABB and SupportHull-only retain IFK-AA+d23. Q3 inherits the RBF build/query backend, collision checker, postprocessor, and data-integrity protocol, but not the Shelf anchors, b100 cap, or Shelf cache root. Its six robot-difficulty profiles use robot-local roots: IIWA and Panda replay depth-23 and depth-20 records, respectively, whereas UR5 materializes endpoint evidence. For cache-backed profiles, Warm Build includes lookup, reads, and demand paging but excludes snapshot creation/opening and the one-time d23 prewarm (16.8 million records, a 12.74 GiB serialized cache, and 1008.6 s); peak resident memory is not reported. The Q1 controls are compound profiles that vary endpoint source, replay, or link representation. Q4 separately measures endpoint construction and isolated link-envelope operations.

A. Q1: Shelf+IIWA Replay and Profile Controls

Fig. 3 shows the fixed cycle through approach-side (AS), target-shelf (TS), center-shelf (CS), left-bin (LB), and right-bin (RB), executed as AS-TS-CS-LB-RB-AS. Table III reports

TABLE V
RANDOM-SCENE RESULTS AT METHOD-SPECIFIC, IN-SAMPLE OPERATING POINTS

Scenario	RBF			PRM			RRT-Connect		BIT*	
	Build	Task/q	$L/L_{ref,\lambda}$	Build	Online/q	$L/L_{ref,\lambda}$	Online/q	$L/L_{ref,\lambda}$	Selected	Online/q $L/L_{ref,\lambda}$
IIWA-Medium	0.075	0.009	1.00	3.567	0.038	1.03	0.002	1.02	0.014	1.05
	[0.058, 0.083]	[0.007, 0.011]	[1.00, 1.03]	[3.007, 3.933]	[0.014, 0.070]	[1.02, 1.12]	[0.001, 0.002]	[1.01, 1.05]	[0.007, 0.042]	[1.03, 1.07]
IIWA-Hard	0.097	0.014	1.00	4.175	0.048	1.03	0.002	1.03	0.027	1.05
	[0.086, 0.105]	[0.011, 0.015]	[1.00, 1.03]	[3.408, 5.637]	[0.021, 0.102]	[1.02, 1.08]	[0.002, 0.003]	[1.00, 1.07]	[0.012, 0.076]	[1.03, 1.07]
UR5-Medium	0.262	0.075	1.05	3.124	0.323	1.02	0.023	1.23	0.026	1.06
	[0.254, 0.270]	[0.020, 0.086]	[1.00, 1.14]	[2.966, 3.418]	[0.085, 0.426]	[1.00, 1.12]	[0.002, 0.049]	[1.06, 1.54]	[0.012, 0.058]	[1.03, 1.14]
UR5-Hard	0.276	0.083	1.05	2.488	0.419	1.06	0.048	1.36	0.046	1.07
	[0.264, 0.284]	[0.043, 0.108]	[1.01, 1.14]	[2.325, 3.048]	[0.129, 0.437]	[1.01, 1.19]	[0.008, 0.090]	[1.14, 1.58]	[0.022, 0.091]	[1.03, 1.17]
Panda-Medium	0.250	0.028	1.05	3.431	0.066	1.06	0.005	1.22	0.016	1.04
	[0.218, 0.265]	[0.013, 0.039]	[1.02, 1.09]	[3.364, 4.180]	[0.038, 0.159]	[1.03, 1.08]	[0.002, 0.011]	[1.12, 1.35]	[0.007, 0.031]	[1.02, 1.06]
Panda-Hard	0.224	0.043	1.05	2.337	0.054	1.10	0.056	1.25	0.062	1.06
	[0.221, 0.237]	[0.021, 0.053]	[1.01, 1.16]	[1.627, 3.978]	[0.034, 0.126]	[1.05, 1.16]	[0.019, 0.136]	[1.14, 1.37]	[0.027, 0.124]	[1.04, 1.12]

Notes: Seconds; medians [25%, 75%]. RBF Task/q: ten-query scene-adaptation-plus-query average; OMPL Online/q: 100 queries/condition. BIT* is selected per instance; UR5 Medium uses query-local edges; postprocessing is excluded.

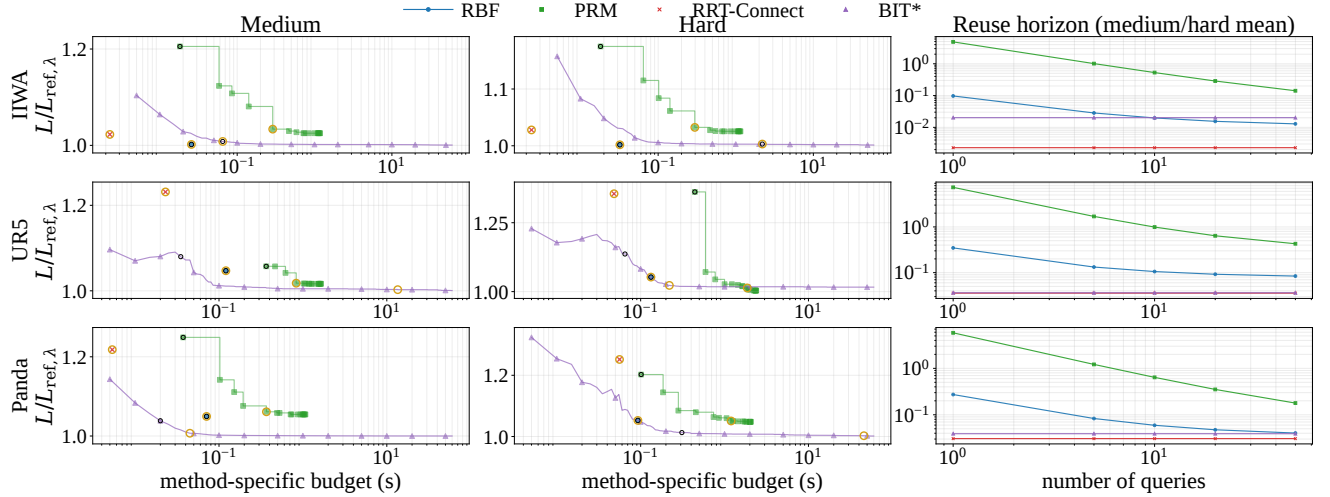


Fig. 5. Random-scene trade-offs. Left: OMPL traces and reported RBF profiles at $K = 5$. Black rings mark each method’s first full-catalog operating point; gold rings mark the reported RBF profiles and the comparison-method points selected by the $\geq 1\%$ aggregate path-length-ratio rule. Right: selected Build/ $K + \text{Task/q}$ projections averaged over Medium and Hard as a function of K ($K > 10$ extrapolated). Table V uses BIT*’s separate 8% per-instance rule and reports query-matched ratios.

the b100 profile over seeds 0–7; b100 was selected during development on this cycle and then held fixed for Q1 and Q2. *Takeaway.* The warm-d23 IFK-AA/SupportHull baseline completes all 40 tasks; the median Build time and within-batch task-completion timestamp are 0.171 and 0.077 s, respectively. HIFK-5 is the registered no-cache contrast. Because it changes both endpoint construction and replay, its 2.770-s Build ($16.2\times$) is not a cache-only effect. Under the registered caches, Critical-sample replays d23 but is non-certifying, so its outputs remain validation-required; Link-AABB and SupportHull-only retain IFK-AA+d23. Table III therefore supports compound-profile comparison, not factor-wise attribution.

B. Q2: Shelf Cross-Algorithm Context

RBF, IRIS/GCS, and PRM use reusable structures configured from the fixed five-query set; RRT-Connect and BIT* run independently per query. PRM shares one endpoint-preloaded cumulative roadmap per seed. Operating points use all five queries and eight runs per method.

For Table IV, PRM and BIT* independently select, for each query, the checkpoint with the lowest median online-

query time among those producing successful saved paths for all eight seeds and a mean path length within 1% of the method–query minimum. IRIS/GCS+post instead selects the fastest construction prefix completing 40/40 tasks with an aggregate ratio within 8% of its minimum. Fig. 4 uses separate method-level 8% display markers; the RBF marker is the registered b100 point. All selectors use the reported Shelf runs, so Q2 compares selector-conditioned path quality, not held-out performance or cross-planner latency. A common postprocessor determines path statistics; its time is excluded. *Takeaway.* Under these method-specific, in-sample selection rules, the median normalized path-length ratio for RBF is below the corresponding PRM and RRT-Connect medians for all five queries, but above the BIT* median for all five and the IRIS/GCS+post median for four. These descriptive path-length comparisons do not establish a latency ranking.

C. Q3: Random Scenes Across Robot Models

Dataset and selection. Q3 uses a frozen catalog of three robots, ten seeds per difficulty, and ten queries per scene: 60

TABLE VI
ENDPOINT-ENVELOPE EVALUATION OVER JOINT-INTERVAL SIDE LENGTHS (RADIANs)

Source	Volume ratio						Time (μ s)						Gap (mm)					
	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50
IFK-AA	1.09	1.23	1.52	2.30	3.53	8.61	5.48	6.11	6.45	6.58	6.81	7.35	–	–	–	–	–	–
HIFK-3	1.04	1.10	1.22	1.51	1.89	3.12	25.94	26.85	29.16	27.21	29.27	29.38	–	–	–	–	–	–
HIFK-5	1.02	1.07	1.13	1.30	1.51	2.07	70.40	71.69	73.82	73.15	75.92	72.32	–	–	–	–	–	–
Critical-sample	1.00	1.00	1.00	1.00	0.99	0.98	8.68	9.04	10.03	11.56	12.69	16.33	-0.05	-0.30	-1.17	-4.53	-10.77	-30.97
Analytical	1.00	1.00	1.00	1.00	1.00	1.00	6429.96	6500.41	6602.25	6800.10	6927.12	7096.01	0.00	0.00	0.00	0.00	0.00	-4.49
Monte Carlo	0.64	0.70	0.74	0.78	0.79	0.82	2251.74	5629.24	11233.26	22352.95	33813.58	56835.57	-6.05	-14.62	-32.68	-65.57	-88.87	-123.53

Notes: 1000 boxes/width; medians; volume ratios use Analytical as the denominator. Gap is the negative largest per-axis width shortfall to the per-box reference extent hull; more negative is worse. Only IFK-AA and HIFK are candidates for the theoretical conservative branch.

TABLE VII
LINK-ENVELOPE EVALUATION OVER JOINT-INTERVAL SIDE LENGTHS (RADIANs)

Envelope	Core volume (m^3)						Build (μ s)						Test (μ s)					
	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50
Link AABB	0.0136	0.0201	0.0384	0.147	0.508	4.61	0.138	0.152	0.167	0.166	0.171	0.178	0.586	0.605	0.601	0.604	0.630	0.790
SupportHull	0.000359	0.00266	0.0143	0.106	0.449	4.54	0.205	0.214	0.218	0.219	0.217	0.250	12.520	12.850	12.267	12.223	12.659	13.026

Notes: Means over 1000 boxes/width and five repeats; no early exits. AABB uses overlap; SupportHull uses direct GJK; Build excludes endpoint construction.

scenes spanning the Medium and Hard settings and 600 saved scene–query instances per method. For each robot–seed pair, the Hard obstacle prefix strictly extends the Medium prefix; queries are shared, and direct start–goal segments are obstructed. The obstacle prefixes and one RBF profile per robot–difficulty condition were tuned and evaluated on this same catalog. Each reported profile completed all 100 instances and was registered from the full-completion candidates after joint inspection of Task/q and path length on that catalog. Q3 is therefore descriptive and in-sample, not a held-out generalization test.

Compared methods. RBF reuses one query-agnostic forest for each ten-query scene. PRM builds one endpoint-preloaded cumulative roadmap per scene; RRT-Connect and BIT* are query-local. The full-success RRT-Connect and PRM curves retain a new point only after an aggregate path-length-ratio improvement of at least 1%. Table V selects BIT*’s fastest per-instance checkpoint whose path length is within 8% of the method–instance minimum, whereas Fig. 5 uses the 1% display rule.

Results. Under these method-specific, in-sample selections, the reported RBF median normalized path-length ratios are 0.02–0.31 lower in absolute ratio units than the corresponding RRT-Connect medians in all six conditions. They are below the PRM medians in five conditions (0.03 above for UR5 Medium) and below the BIT* medians in five (0.01 above for Panda Medium). With Task/q fixed, the arithmetic Build/ K +Task/q projection is 69.6–79.7% lower at $K = 10$ than at $K = 1$ across the three robot-specific averages; this is not a measured multi-query or cross-revision speedup.

D. Q4: Endpoint and Link-Envelope Mechanisms

Tables VI and VII report Q4 over six joint-interval widths. Each study uses a deterministic set of 1000 IWA joint boxes per width, shared across all representations within that study.

The endpoint study reports median construction time and volume. For each box, Gap is the negative of the largest axis-wise envelope shortfall relative to the extent hull of the Critical-sample, Analytical, and Monte Carlo references; more negative values indicate larger shortfalls. This empirical hull is a diagnostic comparator rather than a certificate of conservatism.

The link study reports mean isolated envelope-build and separation costs rather than production-cascade latency. SupportHull’s AABB broadphase and all early exits are disabled. Core volume is the zero-radius centerline-envelope volume B_ℓ° before radius lifting.

Takeaway. At 0.50 rad, HIFK-5 has normalized endpoint volume 2.07 versus 8.61 for IFK-AA (76% lower) at 9.8 times the endpoint-construction cost. Across the tested widths, isolated AABB overlap takes 4.7–6.1% of the mean per-probe time required by direct SupportHull/GJK.

E. Additional Scene-Cache Update Timing

Table VIII records additional scene-cache update timings over eight saved ordered scenes outside Q1–Q4; planning is excluded. Brackets give time interquartile ranges, and ratio ranges give interquartile ranges of per-scene Warm@15/update values.

F. Discussion and Scope

Three components form one algorithmic interface: canonical dyadic addresses couple endpoint replay with forest refinement; exact-union coalescing compresses the current graph without enlarging the certified free set; and typed box and OBB transitions compose corridor witnesses. The dyadic cache hierarchy and region-graph admission remain separately controlled: cache depth bounds replay addresses, the admission cap limits evaluated frontier cells, and anchor coverage affects

TABLE VIII
ADDITIONAL SCENE-CACHE UPDATE TIMINGS (OUTSIDE Q1–Q4)

Operation	Time (ms)	Warm@15/update
Warm build (10 obstacles)	922.75 [897.57, 955.67]	–
Insert 10 → 15	6.06 [5.69, 6.54]	153.3–207.5×
Remove 15 → 10	0.35 [0.31, 1.65]	943.6–3354.0×
Warm build (15 obstacles)	958.79 [941.67, 1115.18]	–

Notes: Eight saved ordered scenes; medians [25%, 75%]. Ratio ranges are interquartile ranges of per-scene Warm@15/update values. Planning is excluded.

only frontier-cell priority. Primitive keys remain unchanged by revision-local unions. Changing endpoint construction changes κ_E , and changing link representation changes δ , but neither changes certificate-status semantics.

Candidate generation is outside the certificate condition. A future, unevaluated integration could expose the validator as an external checker for sampling- or optimization-based planners and learned generators, including vision–language–action (VLA) models such as OpenVLA and world-model–action (WMA) architectures such as UnifoLM-WMA-0 [35], [36]. Such an interface could reuse endpoint evidence, but its end-to-end latency and amortization remain to be measured. In this integration, the upstream planner or model would generate candidates. RBF would retain a complete finite cover of regions validated for the current scene or flag unsupported windows for repair or rejection. Under Assumption 1, only a fully covered candidate would be certificate-backed under Definition 1 and meet the premise of Theorem 1. Developing this interface requires mapping learned action chunks to continuous joint-space paths, synchronizing perceptual scene revisions, extending the conservative validator to task-relevant collision models, and measuring end-to-end checking latency.

RBF targets repeated queries for a fixed robot under compatible scene revisions. Q2 reports selector-conditioned path quality, Q3 reports selected in-sample profiles, and Q1 and Q4 characterize compound cache profiles and isolated envelope mechanisms, respectively. The safety result establishes geometric collision freedom only for paths with a retained finite-cover witness as defined in Definition 1, under a fixed scene revision and Assumption 1. The reported profiles are not evidence of numerical- contract conformance or per-path certificate coverage. The result does not cover dynamics, tracking error, or obstacles that move during execution.

VII. CONCLUSION

RBF turns obstacle-independent kinematic bounds into reusable planning state. A shared dyadic identity maps each endpoint record to a primitive cell of the rebuilt region graph; compositional corridor verification combines current-scene witnesses, and typed transitions expose repair interfaces. The budgeted builder refines canonical cells and coalesces only exact dyadic unions, preserving the certified free set while reducing the revision-local graph. Under the stated collision and numerical assumptions, the resulting finite-cover certificate guarantees collision freedom at every point of a certificate-backed continuous path. The cross-revision conclusion holds only under the matched-replay and fresh-validation conditions

of Corollary 4. On Shelf+IIWA, the warm profile completed all 40 tasks, and its median warm Build time was 0.171 s. At the method-specific, in-sample operating points, the selected RBF profiles have median normalized path-length ratios below the corresponding RRT-Connect medians in all six conditions and below the PRM and BIT* medians in five. With Task/q held fixed, the projected Build/ K + Task/q metric is 69.6–79.7% lower at $K = 10$ than at $K = 1$; this arithmetic comparison is not a measured cross-revision speedup. The analysis establishes the conditional soundness of the cross-revision reuse contract, while the experiments characterize operating costs and selector-conditioned path-length outcomes. Measuring end-to-end speedup across scene revisions and evaluating whether the certificate interface can serve as a low-latency checker for conventional planners and VLA- or WMA-based candidate generators remain future work.

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